



East West University
Department of Computer Science and Engineering
Course: CSE200
Instructor: Antu Chowdhury

LAB 02: Introduction to AutoCAD – Basic 2D Drawing and Editing Tools

Objectives

- Understand and apply basic 2D drawing tools in AutoCAD.
 - Learn how to create and modify simple geometric shapes.
 - Develop precision in drawing using object snaps and editing tools.
 - Gain hands-on experience in using essential AutoCAD commands for technical drawing.
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Topics Covered

- Making Polylines
 - Making Circles
 - Making Arcs
 - 3 Point
 - Start–Center–End
 - Start–Center–Angle
 - Start–End–Length
 - Making Rectangles
 - Move Tool
 - Copy Command
 - Rotate Command
 - Mirror Command
 - Making Fillets
 - Trim Command
 - Offset Command
 - Object Snap
 - 2D Practice Drawing
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Lab Outcomes

After completing this lab, students will be able to:

1. Draw basic geometric shapes such as lines, circles, arcs, and rectangles.
 2. Apply modification tools to move, copy, rotate, mirror, and trim objects.
 3. Use precision tools like Object Snap for accurate drawing.
 4. Create smooth corners and parallel objects using Fillet and Offset.
 5. Combine multiple shapes to produce a complete 2D drawing.
 6. Develop the ability to understand and execute 2D drafting operations efficiently.
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Lab Activities

A. Basic Drawing Tools

1. **Making Polylines:**
A polyline is a connected sequence of straight or curved line segments that form a single continuous object.
 2. **Making Circles:**
Circles can be created by defining a center point and a radius or diameter.
 3. **Making Arcs:**
Arcs are curved portions of circles, drawn using various methods:
 - **3 Point:** Draws an arc passing through three specific points.
 - **Start–Center–End:** Defines an arc using a start point, center, and end.
 - **Start–Center–Angle:** Creates an arc based on a start point, center, and included angle.
 - **Start–End–Length:** Draws an arc using a start point, end, and a specified length along the arc.
 4. **Making Rectangles:**
A rectangle is formed by specifying two opposite corner points to create a closed four-sided shape.
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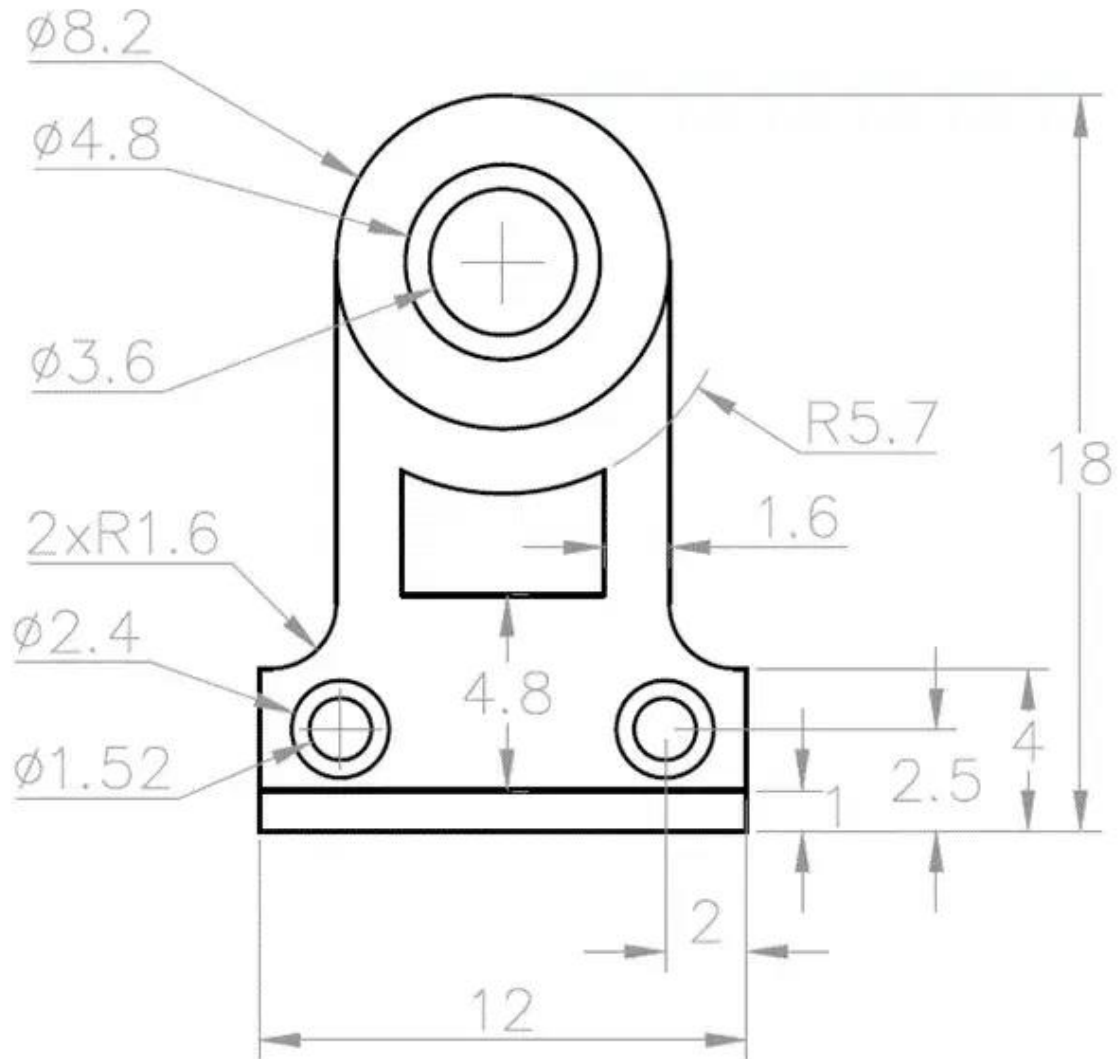
B. Editing Tools

1. **Move Tool:**
Moves selected objects from one position to another.
2. **Copy Command:**
Creates duplicate copies of selected objects at different locations.
3. **Rotate Command:**
Rotates selected objects around a specific base point by a chosen angle.
4. **Mirror Command:**
Produces a mirrored copy of selected objects across a defined mirror line.

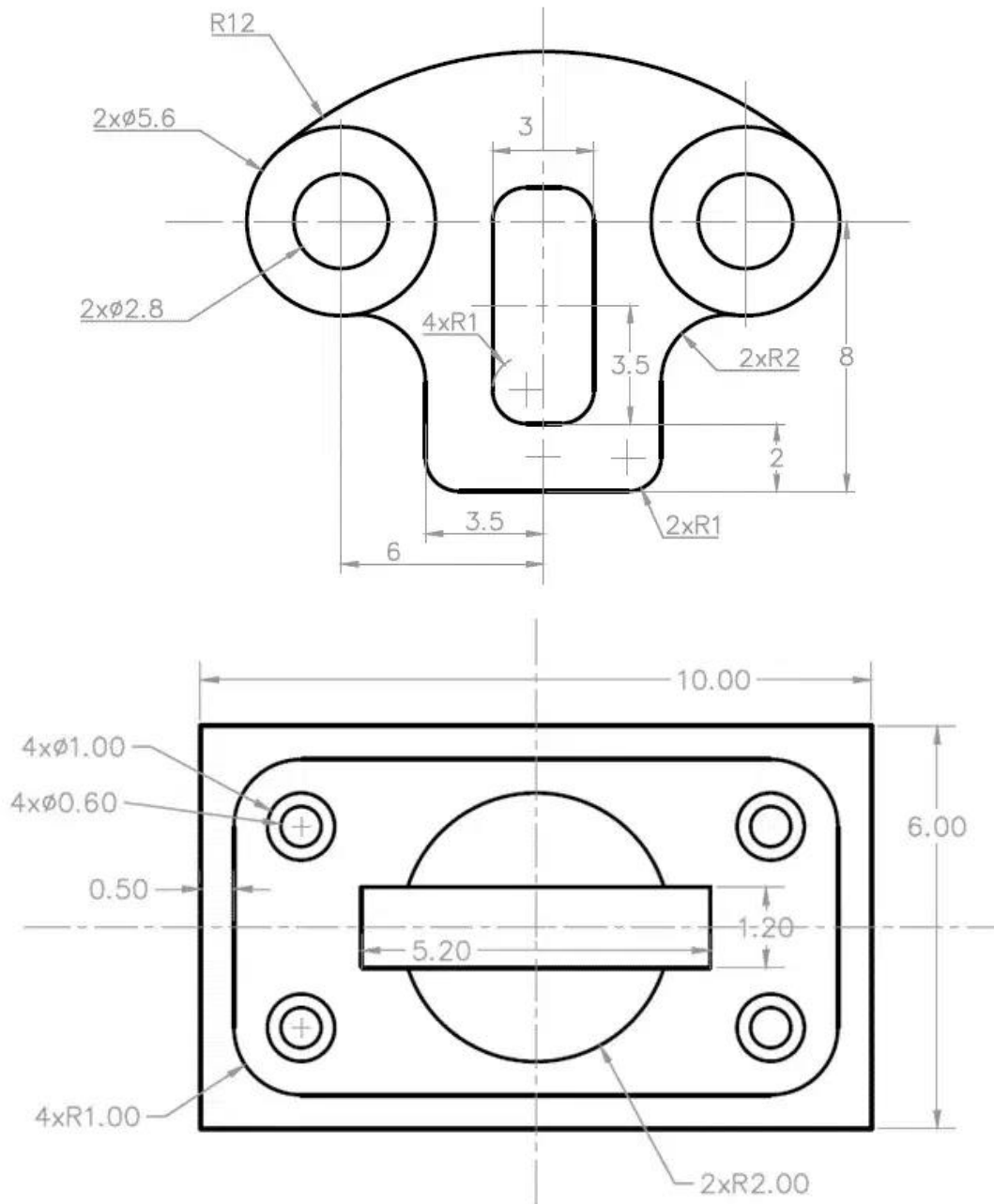
5. **Making Fillets:**
Creates rounded corners between two lines or edges using a given radius.
6. **Trim Command:**
Cuts off or removes unwanted portions of objects that extend or intersect.
7. **Offset Command:**
Generates parallel copies of objects (lines, circles, arcs, etc.) at a specified distance.
8. **Object Snap:**
Helps select exact points (like endpoints, midpoints, centers, or intersections) for precise drawing alignment.

C. 2D Practice Drawing

Using all the tools covered above, create a **2D drawing** attach below:



D. 2D Practice Drawing Assignments



Submission Instructions

1. Complete all drawing tasks in **AutoCAD** and save the file as:
2022-1-60-001_LAB02.pdf (Replace ID with your own student ID.)
2. Submit the PDF file to the designated assignment section in the google classroom.